~~Design High Level~~

Design High Availability Architecture

It can be asked in many other forms like:-

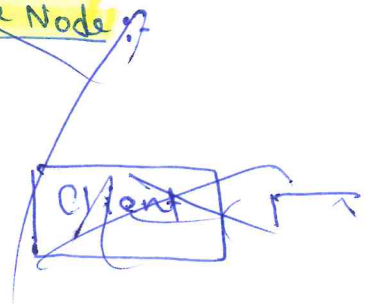
- Design Data Resiliency Architecture. ✗
- Design Architecture to achive 99.999% Availability ✗ (5 9's)
- Design to avoid ~~Singt~~ SINGLE POINT OF FAILURE. ✓
- Active-Passive v/s Active-Active Architecture.

* Resiliency:- Resiliency means capbility to comes out of the failure.

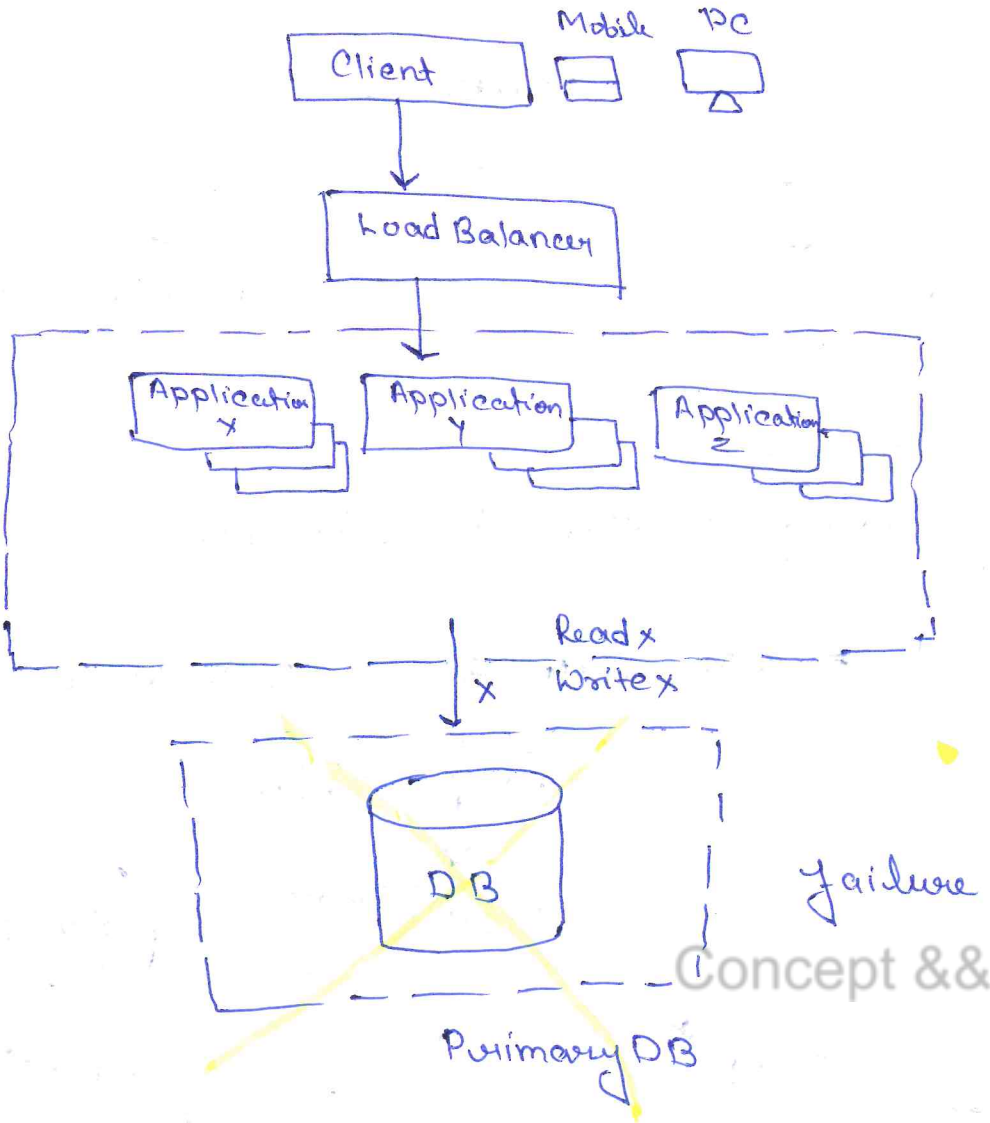
Qu:-
What issue it had why all these things are not achived through Single Node. How we can do it.

Concept && Coding

~~Single Node~~ :-



Single Node :-



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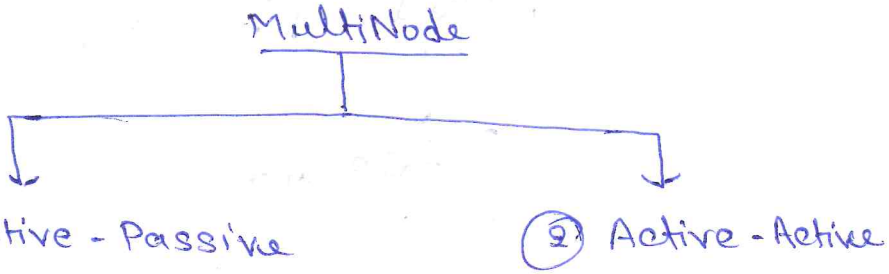
Note :-

- If DB goes down. There is failure happen. What happen now.

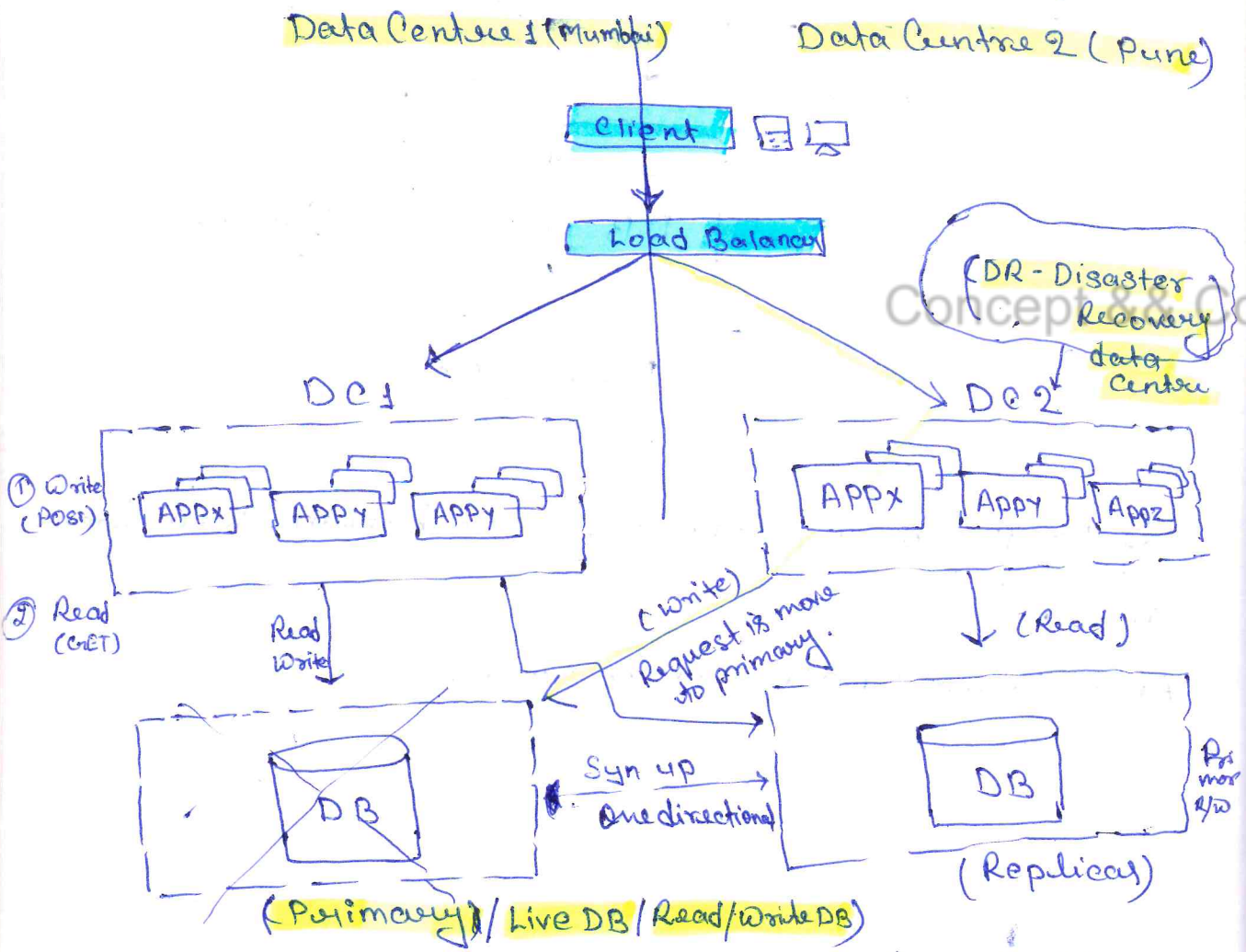
Ans :- Whole Application is down.

MultiNode :- MultiNode has Two Types :



1 :- Active - Passive :→

Every Company has atleast two data Centre.



~~Now~~ Here Active-Passive Architecture only out of two data centres or many more than two also only one can be primary or Live DB or Read/Write DB. Other ones will be treated as Replicas.

Concept & Coding

~~Problem~~:- This is not utilizing resources very well. Because DB2 is free. Why we can't use it?

1. Reason :- Oracle/MySQL/Postgrase are Not Multimaster.

Oracle/MySQL/Postgrea can only write into One primary DB.

Whenever any write (POST) request come

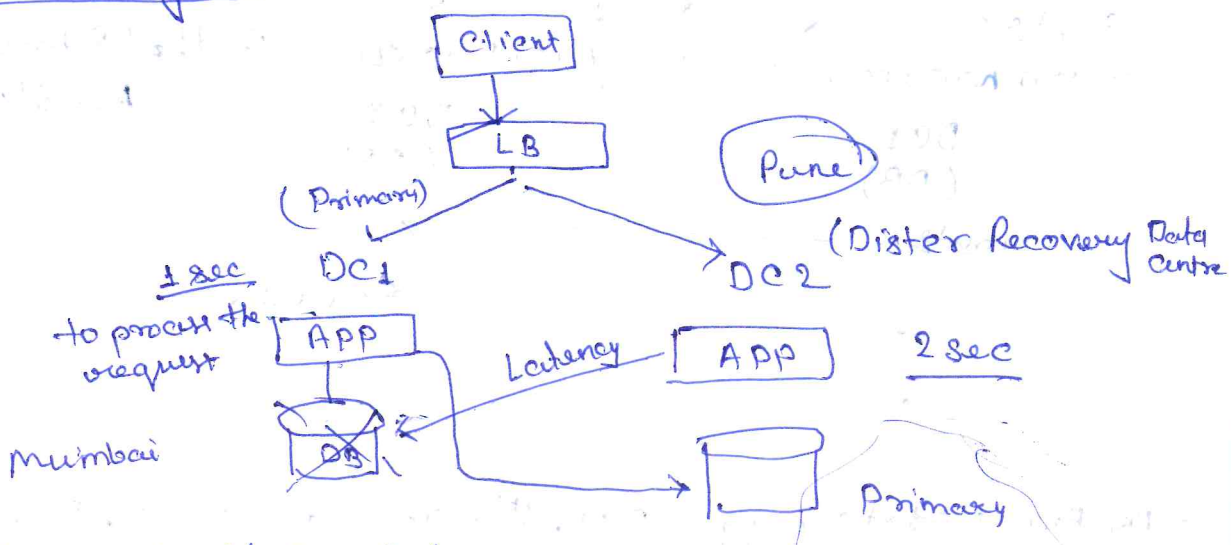
Write (POST) ⟶ Live DB

Qu:- How it is data Resilience and how it is ~~avid~~ avoid SINGLE POINT FAILURE ??

If primary DB Goes down. To easialy sifted to Primary DB to Replicas. we can make replica to primary.

Note:-
If a main DB is down we will route the traffic of that data Center to the One disaster recovery data Center and change it to primary and read write DB.

Disadvantages :-

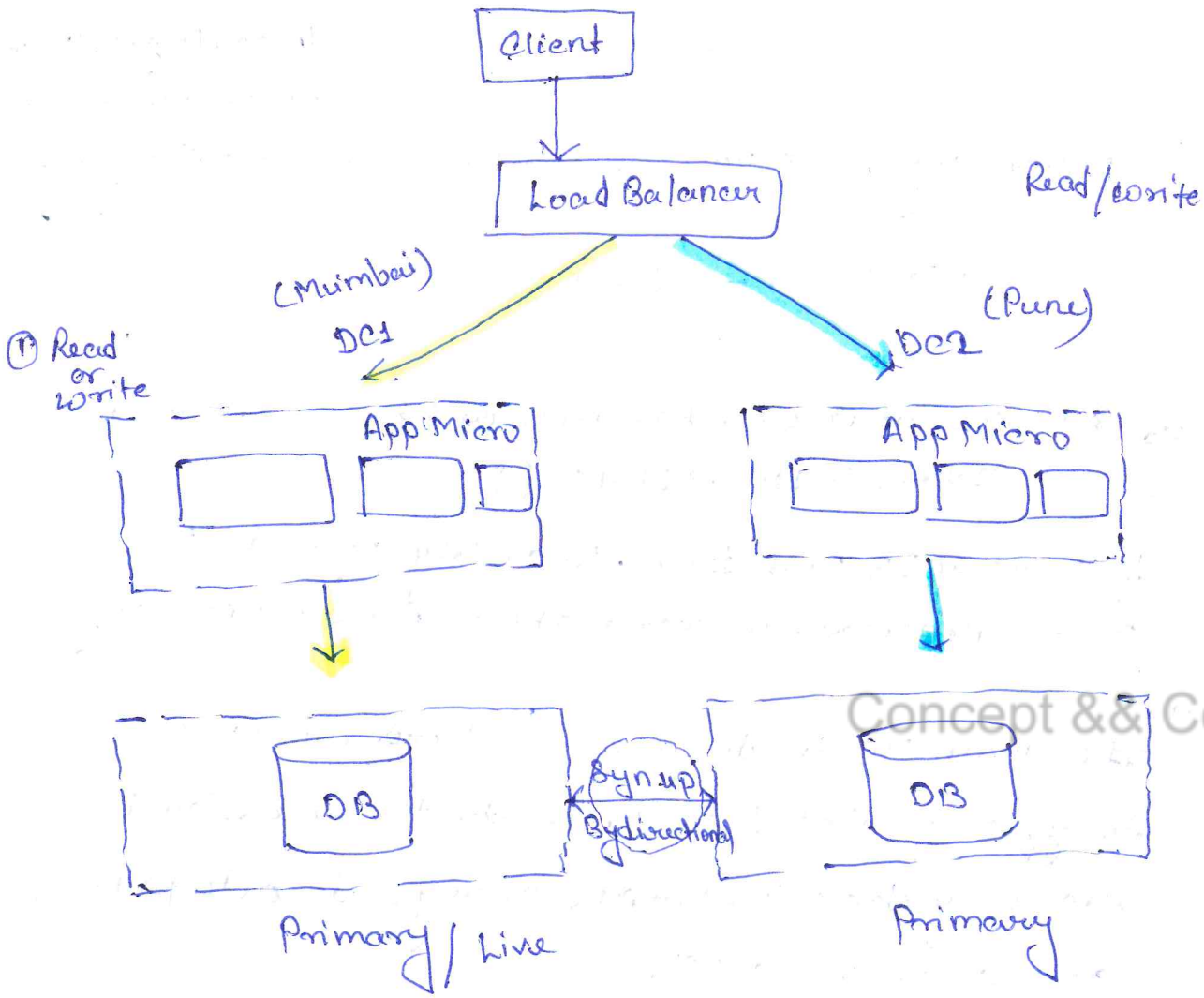


1. There should be latency add on.
2. If DB failure happen the route the traffic on DRD.

Concept && Coding

2. Active - Active Architecture :→

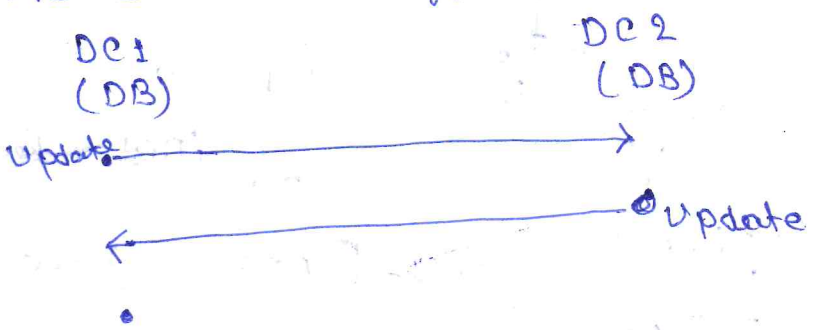
Cassandra - or other NoSQL is supported multimaster.



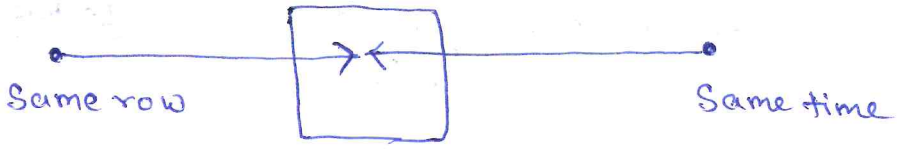
Concept && Coding

Advantages :→

- There is utlizing BB fully. a very well the Data Centre
- It can handle more traffic. Beacuse Batho DB has Read/Write.



- ~~Any~~ Any update happen DC1. It's goes on DC2. DC2 Any update happen It goes on DC1.



- A change can happen on same time and same row. and They are both two replicated each other and There will be conflict.

- There is write happen here on DC1. and Read DC2 ~~Simaltenely~~ samultaneously before even the sync up happen so

* There is a whole lot of things goes in synchronization. There is a lots of deffculties to manage the conflict.

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